Teaching Dossier 2026

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1.0 Teaching Philosophy

Throughout my educational and professional career, my teaching philosophy is guided by four core pillars: **Active and Applied Learning**, **Inquiry-Led Analysis**, **Reducing Barriers to Education** and

Transparent and Intentional Pedagogy. My approach has been guided by feedback from students, mentees, colleagues, and senior leadership teams, ongoing professional development, and an intentional, deliberate reflection about my experiences as an educator. My experience in the classroom, and more recently in academic and intergovernmental agency settings, has taught me that learning is a reciprocal process and is shaped by the diverse experiences, expectations, goals, and prior experience of learners and educators. My approach to teaching therefore strives not only to educate and guide, but also to create opportunities for dialogue, critical reflection, and a shared responsibility in the learning process.

<u>Active and Applied Learning</u>. I have learned to teach with a clear and deliberate focus on engagement and applied learning. My instructional approach has **evolved through experience, feedback, and reflection, leading me to design learning environments that actively involve students** in constructing and applying knowledge rather than passively receiving it. Early on, I recognized that a shift of my teaching and instructing practices from monologue to dialogue in classroom and laboratory settings resulted in more active participation from students. Where I used to sound like the civics teacher in “Ferris Buller’s Day Off” (Anyone? Anyone?), I now prioritized student engagement in their learning by engineering openings for applied learning during lectures, labs, and tutorials. Some examples of the way I have facilitated active and applied learning include student-led recaps at the start of every new lecture/tutorial (to reinforce learning and to develop synthesis and presentation skills), reflection sides during lectures where core concepts were reenforced through quiz-like questions, utilizing contemporaneous digital surveys (e.g. Top Hat, Padlet) for students to ask and answer questions, and soliciting examples of exam or test questions based on that day’s learning. Students who take ownership of course material engage more fully in their own learning, which in turns leads to an enthusiasm for the material, a deeper understanding of the topic, and eventual mastery of core concepts.

<u>Inquiry-Led Analysis</u>. I approach teaching with the understanding that clarity of purpose and relevance are central to student learning. This perspective was shaped by an early classroom moment that fundamentally changed how I design and deliver course material. A student once lamented “I don’t understand how this is even relevant” during an unscheduled classroom break while I wrestled with a temperamental projector. I remember this vividly. I realized then, and upon further reflection, that I had made three key mistakes in developing that day’s lecture material. First, I failed to recognize that not everyone found these topics as interesting as I did, and I could not rely solely on my enthusiasm to carry interest through a 50-minute lecture. Second, I did not do a good enough job of relating the material back to their everyday lives. What I was teaching was new and theoretical, and they had no reality-based frame of reference where they could connect with the content. Third, I did not do a satisfactory job of placing this material within the wider context of the course or their overall learning objectives, and therefore the ‘relevance’ of the material was lost. With experience, I have come to recognize that student engagement depends as much on relevance and context as it does on content expertise. Some of the most influential moments in my development as an educator have emerged from listening closely to students’ responses, both explicit and inadvertent.

I also recognize that not all students I teach will go on to use GIS, remote sensing, geospatial, or geomorphological analysis directly in their future careers. For these learners, my goal is not mastery of specific software or techniques, nor the rote memorization of facts, figures, and configurations, rather, the development of durable, transferable skills. In my lectures, I emphasize critical thinking, clear communication, and independent and group learning. By engaging students in interpreting spatial data, justifying methodological choices, and communicating uncertainty and limitations, I encourage students to think analytically and reflectively. These skills remain as valuable for those individuals whose careers take them away from these scholastic domains, as much as it does for those interested in careers that rely on geospatial analysis.

Reducing Barriers to Education. I believe that my duty as an educator is to make the material I teach as accessible and inclusive as possible by intentionally identifying and reducing barriers that could hinder student success. While reflecting on my own classroom and teaching practices, I have identified three major barriers to student success: **student wellbeing, cultural diversity, and socioeconomic status**.

Student Wellbeing. I have witnessed student wellbeing manifest as low confidence, a lack of sense of belonging in certain academic settings, overall cognitive load, misunderstanding course expectations, and general mental wellness. I work to address and dismantle these barriers by designing course material with care and embedding collaborative learning activities that foster peer support and shared responsibility for learning. I also solicit feedback from students early in the course to ensure that the deadlines I have laid out do not, within reason, conflict with major academic milestones from other classes. For example, in one laboratory section I led, almost all the students were enrolled in another class which had a major project due at the same time a relatively routine lab report was to be handed in. Working with the course instructor and the students, we found a mutually beneficial solution to this overscheduling that students later said had a positive effect on their stress levels during that period.

Cultural Diversity. I am acutely aware that culture imbues every aspect of teaching and learning, and to fail to account for diversity in values, ethics, communication styles, and interpersonal relationships can ultimately compromise the learning process. Having lived and worked abroad since 2020 within highly international academic and research environments, I bring this awareness directly into my teaching practice by intentionally using inclusive language and avoiding assumptions about learners' identities, experiences, or prior knowledge. In my teaching practice going forward, I endeavour to design learning spaces that reduce reliance on unspoken norms and support multiple ways of engaging. I establish clear expectations for respectful communication, offer varied modes of participation beyond spontaneous verbal contribution, and structure collaborative work with explicit roles and guidance to promote equitable participation. The students I have had the privilege to teach, and will continue to educate in the future, come from a wide range of cultural backgrounds, lived experiences, and personal identities and perspectives; it is my duty to ensure all students feel safe, respected, and valued, regardless of their positionality.

Socioeconomic Status. I am cognizant of the ways in which students' socioeconomic circumstances shape educational experiences and access to learning opportunities. Financial constraints, competing work or caregiving responsibilities, access to technology or software, and prior educational inequalities all influence how a student engages with coursework and participates in their learning. This can be especially true in quantitative and technically demanding fields such as GIS and remote sensing. In my teaching practice, I strive to mitigate these barriers by designing flexible and transparent course structures, prioritizing low-cost, open access, or institutionally supported learning resources, and offering alternative ways for demonstrating learning, where appropriate. I am intentional about setting clear expectations, providing scaffolded assignments, and providing students with timely, actionable feedback, which helps to build confidence in the course materials. By remaining attentive to the material realities of students' lives, I am to create learning environments that are equitable, supportive, and responsive, allowing students to succeed regardless of their socioeconomic background.

To address these, and any other barrier a student might face, I believe it is part of my duty to students to be knowledgeable about the university's resources, financial supports, advising services, health services, and student programming so that I can appropriately guide students towards resources that support their well-being, and their learning. And while I recognize the limits of my role as instructor and educator, I consider this to be as important to my teaching pedagogy as any other aspect of my philosophy.

Transparent and Intentional Pedagogy. My pedagogical approach is grounded in transparency and continuous reflection. In the classroom, lecture theatre, lab, or at a field site, my objective is to always ensure students understand the *why* of what we have set out to do. In practice, this can look like intentionally articulating the purpose and relevance of course activities; contextualizing concepts and assignments within the broader disciplinary, social, and professional context; and tying learning outcomes back to course objectives. This transparency encourages student ownership over their learning, deeper motivation, and broader engagement.

My teaching is further shaped by a deep commitment to reflection. I regularly solicit formative and summative feedback from students, colleagues, and academic leaders, and I engage with evidence-based instructional practices and professional development opportunities to continually refine my pedagogy. And when opportunities for formal feedback was missing from established procedures, I created my own opportunity to learn. Specifically, I developed an **Invited Lecture Feedback Form** to collect assessments of my teaching and lecturing approaches. Further explanation about the Invited Lecture Feedback Form can be found in Section Existed and Appendix A. I view teaching and educational leadership as dynamic, iterative practices that require ongoing feedback, self-reflection, and growth.

Final Thoughts. My teaching philosophy is grounded in intentionality, inclusivity, and continual growth as an educator and a leader. To me, teaching has evolved from a one-sided one-size-fits-all mentality to a vibrant reflection the practices of active learning, inquiry-led analysis through the dismantling of institutional barriers, and transparent and intentional pedagogy. I strive to foster learning environments that invite inquiry and participation, while connecting fundamental topics to real-world problems. I am committed to developing my pedagogical approaches to support diverse learners in diverse learning environments to equip students with the analytical, technical, and transferable skills they need to succeed both in the classroom and out in the real world.

2.0 Teaching Experience

My academic teaching responsibilities have included **classroom instruction, laboratory and field demonstration, and supervision of student research**. Below are several examples of those responsibilities, and how I approached them.

2.1 Academic Lecturing Experience

Classroom Instruction at the University of Calgary. While enrolled as a PhD student then Candidate at the **University of Calgary**, I was invited to guest lecture on numerous occasions for diverse sets of topics. Below are examples of individual lectures and lecture series I was invited to develop and present.

Geography 201: Physical Geography (2019)

In this lecture series, I lectured to two sections of this large first-year class on the topic of planetary geomorphology. I specifically tailored my lecture to address the evolution of Mars from a wet, warm world through to the frigid, arid planet we know today. I utilized power point slides with review slides at the end of every major topic (there were 3 sections to this lecture), videos, and mini quizzes throughout the presentation to keep students interested and engaged.

Geography 407: Wind Science (2017)

In this lecture series, I delivered 3 complementary, yet distinct lessons: *Wind Interaction with Mountain Topography*, *Aeolian Processes: An Introduction*, and *Aeolian Processes in our Solar System*. Within this series, I presented progressively more complex topics and concepts over the week I lectured. Following this series, I asked students to provide me with feedback on my presentation and delivery of material (Appendix B). This feedback allowed me to tailor my presentations for subsequent lectures outside this class.

Between 2015 and 2018, I was also invited to lecture on 3 more occasions, discussing topics similar to the lecture series I developed in 2017.

"Engaging and Enthusiastic, really enjoy your lectures. Good when you draw it back to your research in the end" – Student Feedback, Geography 407

"You can tell she's really enthusiastic about the material and is really knowledgeable the content and can answer any question posed at her" – Expert of student feedback, Geography 407

I was invited to guest lecturer in the following courses:

- ***Geography 415: Hydrology (2017).*** *So You Want to be a Hydrologist? Experiences from the Arctic*
- **Geography 205: Gateway to Geography (2018).** *Atmospheric Hazards: The Anatomy of Cyclones*
- **Geography 437: Cartography and Geographic Visualization (2018).** *A Cartographic Trip Through the Solar System*
- **Geography 280: Thinking Spatially in a Digital World (2020).** *Applying GIS Techniques to the Study of Mars*
- **S808: Space Science, The Open University (UK) (2022, 2023).** Guest Lecture and Mars Yard Facilitator. *The Physics of Sediment Transport on Earth and Mars*
- **S283/284: Planetary Science and the Search for Life/Astronomy, The Open University (UK) (2023)**. *Planets and Moons Chat (https://tinyurl.com/webcastOU)*

Classroom instruction at Queen's University. During my time as a Master's student at **Queen's University**, I was invited to lecture on diverse topics for second- and third-year courses. Master's students were historically not allowed to lecture to undergraduate students, **but an exception was made for me**, as I had sought out the opportunities and demonstrated that I was eager to develop as an educator.

Geography 208: Surface Processes, Landform, and Soils

In this lecture series (3 lectures), I developed introductory material about glacier physics, permafrost, and Arctic soils for a second-year class. I was mentored by the course instructor on how to structure the lessons, as well as the key topics that were to be covered.

Geography 312: Watershed Hydrology

For these series of two lectures and one laboratory instruction, I was tasked with developing material to engage students in watershed hydrological modelling. This was the first opportunity I had to create my own teaching and laboratory material.

Guest Lectures, Invited Seminars, and Outreach Presentations. Since being awarded my PhD, I have continued to lecture for colleagues in Canada and Europe. Below are some of the lectures, public seminars, and outreach activities I have been invited to present at:

- **Guest Seminar, Basel University (Switzerland)**. *The Aeolian Environment at Oxia Planum. Supporting Rover Operations through Terrestrial Analogues, Atmospheric Modelling, and high-Resolution orbital observations.*
- **Outreach Lecture, Galway Geological Association (UK)**. *The Mysteries of Mercury and Mars, revealed.*
- **Guest Seminar, South East Physics Network (UK).** *Modern Mars: A (Brief) Exploration of a Dynamic World.*
- **Guest Seminar, International Women and Girls in Science Day (UK).** *My Journey as a Woman in Planetary Science*
- **Guest Seminar, Rover Science Operations Working Group (Remote).** *The Aeolian Environment of Oxia Planum: What Rosalind Franklin Might Find (and why it's important we look).*

- **Guest Lecture, GRADNet Astrobiology and Planetary Science Workshop (UK).** *The Physics of Sediment Transport on Mars (And Why We Should Care).*
- **Invited Outreach Talk and Activity, Calgary Area Girl Guides of Canada STEM Day (2018-2022)**. *Various outreach talks and activities (cratering experiments, rover simulations, bottle rocket launches, online activities during Covid, etc.) designed for children ages 5-12.*

2.2 Academic Laboratory, Tutorial, and Field Experience

Geography 403: Oceanography (Formally Oceanography and Climate Variability) (2015, 2016, 2018)

I was the sole Teaching Assistant in this course. I was responsible for delivering weekly labs, which consisted of traditional science laboratory analysis of datasets, as well as written and oral assignments. I developed and presented introductions to each lab which provided students with an overview of the day's assignment, as well as a brief review of the materials I had handed back. Student feedback can be found in Appendix C.

For this course, I developed an introductory Excel module (see Section 3.1 Established Teaching Practices (What I've done so far)) to assist students in their data analysis. I was also given the opportunity to create a lab exercise in 2015 (Appendix D), which students completed as an optional assignment (all students that year opted into the activity). The success of this exercise was such that it was then adopted by the Instructor of Record for future offerings of the course.

"Elena approaches this position with uncommon enthusiasm. She has been very available for additional help and provides continuous feedback. One of the best T.As I've had!" – Student Feedback, USRI, Geography 403, 2015

"Without a doubt, one of the very best T.A.s in the department. Would highly highly recommend her for a T.A. awards. Will try to take courses with her again" – Student Feedback, USRI, Geography 403, 2016

"Elena did an excellent job this semester. She worked independently, and delivered all of the lab assignments without any problems…I basically didn't hear anything from the students. Typically that's a very good sign, because if things are not going well in the lab students are usually pretty quick to bring their issues to me" – Dr. Brent Else, Graduate Assistant Performance Review of Elena Favaro, Geography 403, 2018

Geography 415: Hydrology (2017)

I was one of two Teaching Assistants assigned to course. Along with my colleague, we delivered five traditional science laboratory assignments by each being solely responsible for two and delivering the final lab together. The lab component of this class was viewed as a supplement to the in-class lectures. As such, I was responsible for the development and delivery of introductory lab materials for which the students were responsible for.

Geography 391: Field Methods in Geography – Geography Field School (2015)

I was one of several Teaching Assistants assigned to this course. I was responsible for teaching students how to navigate terrain and read maps – skills they would need to complete the modules of the course. I also lectured on the topic of regional geomorphology. I was also responsible for student supervision and safety and drew on my extensive experience and certifications to provide instruction on backcountry safety prior to field school.

Below are the courses I was a Teaching Assistant for at **Queen's University** between 2011 and 2014. I was responsible for laboratory and tutorial facilitation for each course.

- Geography 102: Earth System Science
- Geography 208: Principals of Geomorphology and Pedology
- Geography 205: Geography of Canada
- Geography 304: Arctic and Periglacial Environments
- Geography 312: Watershed Hydrology

2.3 Academic Supervision Experience

In my role as Postdoctoral Research Associate at the Open University, as well as Research Fellow at the European Space Agency, I have had the pleasure of supervising undergraduate and graduate research in both informal and formal capacities.

Most recently, **I supervised a student researcher** hosted at ESA's ESTEC site in the Netherlands through the University of Leiden's LEAPS (Leiden/ESA Astrophysics Program for Summer Students) initiative. After submitting and being awarded a research project proposal (broadly, investigating traversability paths for the ExoMars Rosalind Franklin rover), I led the full selection process, including shortlisting, interviewing, and hiring the student. I provided training in introductory Martian geology and geomorphology, as well as the GIS and remote sensing datasets, software, and analytical tools required to address the project's scientific objectives. Under my guidance, the student became proficient in identifying and using remote sensing data from multiple NASA and ESA missions, applying advanced geospatial techniques to both primary and derived datasets, and developing an ArcPy-enabled processing pipeline that streamlined a complex analytical workflow. The results have been extremely well received by ExoMars program scientists, engineers, and planetary researchers. This work will be presented at several upcoming European scientific conferences, and a manuscript detailing our methodology and findings is currently in preparation. The approach we developed shows promise for supporting strategic science operations during the nominal ExoMars Rosalind Franklin rover mission, and its algorithms will be evaluated in upcoming rover simulation campaigns.

I currently supervise a **graduate student research intern** to ESTEC, whose work focuses on building a lightweight, site agnostic terrain classification algorithm for the rapid reconnaissance of high-resolution images of the Martian surface. I am responsible for instructing the student on Mars geology and geomorphology, as well as GIS and remote sensing fundamentals. It should be noted that intern supervision at ESA is a competitively-awarded process; my initial research proposal went through a rigorous internal review process, and I have led the full selection process.

2.4 Non-academic Teaching Experience

Outside of academic pursuits, I have a substantial background in non-academic coaching and teaching.

I coached squash for children aged 4-12, for which I obtained formal coaching certifications. I developed age- and skill-appropriate training sessions that prioritized fun while emphasizing skill development, confidence, and a sense of teamwork in an individual sport.

I voluntarily completed a two-year qualification program to become a certified Trainer with the Girl Guides of Canada. In this role, I taught adults in both structured classroom settings and practical field environments. I delivered mandatory trainings and voluntary continuing education courses in classrooms combining lecture-style presentations, hands-on learning, collaboration, and gameplay. In the field, I taught practical camping and wilderness skills. After moving abroad, I continued to train Leaders with Guiding UK.

Across these roles, I gained **experience in curriculum design**, **experiential learning**, and **adapting teaching methods** to diverse groups and learning styles.

3.0 Teaching Methods

3.1 Established Teaching Practices (What I've done so far)

Creating opportunities for engaged learning takes different forms depending on class size, the duration of my teaching commitment (a guest lecture verses a teaching assistant), and the location of learning (in a lecture hall, a computer or dry lab, in the field, and remote and in-person).

The Scaffolding Approach. When I guest lecture, I employ a **scaffolding approach of 20/60/20** to presenting material. Often, the topic I am teaching is based on my own research interests and academic specialty. In order to ensure all students can engage with the material and follow the lecture, I start by introducing core concepts that will be necessary to understand the more technical or challenging information to come. I devote 20% of the lecture time to this first platform of the scaffold. The next 60% consists of a more "traditional" lecture format, and the final 20% connects all the lecture material to a real-world application, ties it back to learning outcomes, and/or places it within the context of the wider course. Regular opportunities for active learning **(structured interruption)** occur at the end of each platform section, when I reinforce the material by providing skeleton slides with words missing from key summary statements we fill out as a class, and/or I administer a short online quiz (≤ 5 questions) to assess comprehension. This technique gives me real-time information about the kind of material that is well understood, or concepts I need to revisit before moving on. When I present a **series of lectures**, the scaffolding ratios change to reflect the feedback I received in previous lectures.

Reactive Learning to Proactive Instruction. During my educational journey, I observed that many students have not developed database management skills essential to success in the class, laboratory, or in their overall degree programs. In response, I have intentionally integrated skill-building resources into my tutorials that support the immediate course/laboratory objectives but also support students' long-term academic and professional development. For example, in

Geography 403 (Oceanography), students were tasked with working with multi-parameter datasets to complete a laboratory assignment. During that exercise, I found students were not asking questions about the exercise material, but rather how to analyze and visualize these data in Microsoft Excel. I had encountered similar situations in other classes as well. To address this issue for my own students, I developed interactive Excel tutorials that walked students through foundational expressions and analyses step-by-step. I incorporated these tutorials in other course and laboratory assessments I was responsible for and shared them with other teaching assistants in the Geography, Biological Sciences, and Anthropology departments to use and modify.

"She was great and often went beyond the scope of her TA expectations and promoted skills and resources that would help us beyond this class" -Student Feedback, USRI, Geography 403

"Friendly and approachable, great with excel software and database files" -Student Feedback, USRI, Geography 403, 2015

"Thank you SO MUCH for all the excel help. I was panicking about another lab and used all your tips and tricks! Best TA EVA" – Student email, Geography 403, 2018

"Hey, thanks for the excel tutorials. I used them for that fish assignment [Department of Biological Sciences course], and the class found them helpful. I'm also glad I didn't have to field a million questions about how to format a second axis" – Teaching Assistant in a second-year Biological Sciences course.

Development of Video Materials (pre-Covid era). In the summer of 2016, I was asked to develop an interactive exercise for a course on 3D Technologies and Digital Mapping that introduced students to ground-based photogrammetry (the art and science of creating three-dimensional models from photographic measurements). The exercise was designed to guide students through the full 3D modelling workflow. I quickly recognized that the significant computational time required and the potential for software issues would greatly outpace face-to-face time I would have with students. To address these constraints, I developed an online tutorial to support and supplement in-class instruction. Although the course was ultimately not offered, I shared the tutorial publicly on YouTube[1] in October 2016; it has since received over 36,000 views.

The success of that video, and the fun I had creating it, lead me to develop more video tutorials, often because I found a methodology that I felt could help someone else. Tutorials that have been developed but not posted are mostly introductory ArcGIS Pro workflows (buffers, spatial joins, mosaicing datasets, raster calculations) I created for specific purposes. These videos have been shared with students and colleagues but require further clean-up for a general audience. Those efforts are ongoing.

[1] https://www.youtube.com/@efavaro

3.2 Developing New Approaches

In the future, I look forward to experimenting with different methods of delivering course materials and laboratory assignments. Some examples of these approaches are described below.

Active and Applied Learning in Traditional Lecture-style Teaching. In circumstances where traditional classroom lecturing is deemed the most appropriate method for teaching, I look forward to incorporating a range of active and applied learning methods to promote student engagement and to collect real-time and asynchronous feedback on student' understand of course materials. These methods include:

Minute Papers: Ideal for both large and small class sizes, I provide a short prompt at the beginning of the lecture, and students have 60 seconds to write down everything they recall from the previous lecture. They can either keep this for their own edification, or, in smaller classes, they can be handed in for non-credit evaluation and feedback. *For synchronous class formats.*

Student Syntheses: Ideal for small class sizes, students are required to be prepared to briefly (1-2 minutes) recap the previous lecture's core themes and major take-aways. Any student can be called on, which encourages students to synthesize the previous lectures materials prior to the start of a new class. *For synchronous class formats.*

Clear as Mud: Ideal for both large and small class sizes, students anonymously submit one (or more) concept or idea that is unclear, and I can either synthesize results outside of class and return to the next lecture prepared to go over those identified concepts, or set aside time during the lecture to review the results and address those topics in real time. *For synchronous and hybrid class formats.*

Two-Stage Polling (think, vote, discuss, revote): Ideal for both large and small class sizes, a question is proposed and students vote. The result of that vote is broadcast in real time, and then students turn to their neighbour and discuss what answer they chose. All students then vote again, and before/after results are discussed. *For synchronous class formats.*

Exam Question Banks: Ideal for both large and small classes, students can collaborate with their neighbour, or in small groups to craft an exam question they might expect to see based on that day's lecture. Questions can be submitted and results acknowledged in real time. *For synchronous and hybrid class formats.*

Lecture Exit Polls: A final question is posed at the end of lecture that is crafted to gauge a student's comprehension of an aspect or overall theme of that classes' material. The student must answer the question and indicate their confidence in their answer (for example, high, medium, low or very confident, confident, not confident). This will help guide the introduction material for the subsequent lesson. *For synchronous and hybrid class formats.*

Post-Lecture Quizzes: Following the lecture, I would post a voluntary quiz online and ask students to complete it at least 72 hours before the next lecture. This provides me an opportunity to gauge how well the themes of the previous lecture were understood, and will inform how the beginning of the next lecture will be structured; am I ok to do a brief recap and move on, or do I need to spend time on a topic that students need more instruction with? *For synchronous and hybrid class formats.*

"Great enthusiasm, really good job keeping the class engaged with discussion and questions, interesting hands-on activity to end class" – Student Feedback, Geography 437

"…I love her summary slides too, if you miss something during the lecture or not quite get something, the summaries really help with that…I liked the guessing which planet slides for the solar system lecture!" – Excerpt of student feedback, Geography 407

Non-traditional Teaching Methods. For courses where a non-traditional lecturing and laboratory teaching techniques can be employed, I look forward to developing courses that utilize the Flipped Classroom and Problem-Based Learning (case studies) approaches. Brief explanations of how I would incorporate these methods into my teaching are as follows:

> *Flipped Classrooms:* In my flipped classroom, I would provide students with short (10-15 minute) pre-recorded videos or readings that would allow students to learn foundational material at their own pace, prior to attending a lecture. In class, I would follow the method of Mazur (2009)[2] and alternate my teaching around **two-stage polling** multiple choice questions, class discussion, and presentations that reinforce the learning students have just engaged with. Some faculty in the Department of Geography have employed this method, and I would also look to them for guidance on how to effectively employ this active learning approach to my course design. Online resources like the Harvard Kennedy Shool's SLATE (Strengthening Learning and Teaching Experience) toolkit[3], and the Taylor Institute for Teaching and Learning's database of resources, which, at the time of writing, includes nine online resources for flipped classrooms, will also be important to consult when I design courses around this teaching method.
>
> *Problem-Based Learning for Hands-on Laboratory Assignments:* Developing engaging, effective geospatial laboratory activities will be critical for student development, application of course principles to real-world problems, and overall critical thinking and analysis development. I look forward to **developing problem-based exercises for GIS and remote sensing assignments**. I would pose a complex, real-world problem to the lab section and encourage students to break out into smaller groups for discussion. I would ask those groups to identify which class concepts might be needed to consult for this question, what geospatial information they may need to source, and what analysis techniques they might employ. The groups would then be tasked with outlining their methodologies and/or developing potential workflows. The flexibility of implementing this method offers a wide range of opportunities for learning at all stages of the course for students in both small and large laboratory sections.

These methods are only some of the ways in which I plan to encourage active learning and participation in my classroom and during laboratory assignments. I look forward to consulting with faculty in the Department of Geography to understand what has and has not worked in their own classrooms and incorporate their feedback when developing curricula and hands-on assignments.

[2] Mazur, E. (2009). Farewell, lecture? *Science*, *323*(5910), 50-51.
[3] Harvard Kennedy School. (2016). Flipping Kit. Harvard Kennedy School SLATE. https://slate.hks.harvard.edu/flipping-kit

3.3 Ongoing Pedagogical Development

Feedback from students and teaching assistants will be key to developing my methods of teaching and active learning. Partway through the course (perhaps 3-5 lectures into the semester) I would collect informal feedback from **students in class** to gauge how their learning is progressing, how my teaching is resonating with them, and to identify changes I can implement to foster the most fun, effective, and welcoming learning atmosphere possible. Some questions I could ask would be "What do you like about the course so far", where I would list options like "In class activities" (referencing the active learning methods laid out in the section above), "class organization", "clarity of expectations", "slide/material content", etc. Another question could be "Are there any changes you would like to see implemented to make the course a better experience for you?". I could provide options like "No", and for a "Yes" response, allow students to write-in what they would like to see changed, modified, added, or clarified.

I would also speak with **teaching assistants** to understand how laboratory learning is progressing. I would hope that these conversations would happen naturally, as I would continue to check in with those graduate students, but I would also schedule meetings or tag-ups to sit down and discuss how they feel instruction is progressing and their views on whether students are engaging with materials or if a rethink of approaches is necessary.

The next section of this dossier outlines how I have engaged with opportunities to improve my teaching skills and develop my personal pedagogy during the past decade, and how I continue to place a great deal of importance on continuing education.

4.0 Commitment to Ongoing Pedagogical Development

Ongoing learning is central to my pedagogical practice. This means engaging with both foundational scholarship and emerging research, as well as seeking out **professional development opportunities**.

4.1 Teaching and Stewardship

Pedagogical Development: During my PhD at the University of Calgary, I completed the **Graduate Student Teaching Development Badge Program** (now called the Postdoctoral Scholar and Graduate Student Certificates in University Teaching and Learning) through the Taylor Institute for Teaching and Learning (the term "Graduate" identified it as the courses offered to graduate students, while the content of the program was structed around the teaching of undergraduate students). The program consisted of five badge programs, each involving 12 hours of in-person instruction, applied assignments, and a capstone project:

1. Learning Spaces and Digital Pedagogy Program: Graduate Students
2. Theories and Issues in Postsecondary Teaching and Learning: Graduate Students
3. SoTL Foundations Program: Graduate Students
4. Emerging Teachers Development: Graduate Students
5. Developing Your Teaching Dossier: Graduate Students

This program provided a rigorous foundation for my pedagogical development and offered the mentorship, structure, and guided practice I needed to grow as an educator.

I have continued to engage in professional development opportunities to deepen my pedagogical knowledge and maintain momentum in my teaching practice. While at the Open University, I enrolled in the following continuing-education modules (8 hours of instruction, and weekly assignments):

- Introduction to EDI (Equity, Diversity, Inclusion) in Teaching and Learning
- Inclusive Curriculum Development and Delivery
- Being an EDI Change Agent

I am currently enrolled in the first of a three-course module through OpenLearn called "Inclusive Leadership". These courses are challenging me to consider how leadership manifests itself in professional education situations. I have completed over 30 hours of continued education. A list of trainings can be provided upon request.

Together, these programs and workshops highlight my belief that teaching requires a continual investment in learning. I am committed to developing as both an instructor and a mentor, and **I actively seek out opportunities that challenge me to grow, broaden my skills, and deepen my awareness of student needs**. This dedication to self-improvement continues to shape my teaching philosophy and practice.

Student Stewardship: I have completed specialised trainings to support student wellbeing:

1. ASSIST: Applied Suicide Intervention Skills Training
2. Suicide Prevention Training
3. Responding to Students in Distress
4. Bystander Awareness Training

These trainings did not necessarily inform my teaching practices, but it helped in raising my awareness about what kind of behaviours, patterns, and mannerisms students (and indeed, anyone) might exhibit while in acute distress.

4.2 Creating Opportunities for Feedback Where None Existed

To collect feedback about my teaching approaches from students who attended my guest lectures, I created an **Invited Lecture Feedback Form** with help and resources from Dr. Carol Benson at the Taylor Institute for Teaching and Learning. I distributed these feedback forms for classes in which the instructor of record allowed me to do so. The comments I received helped to shape and refine my pedagogical approach to classroom lectures. Other graduate students across the university (in Anthropology, Kinesiology, Biological Sciences, Physics, Social Work, Geology, English, Film Studies, and Linguistics [please note that these are how the departments were named when I was enrolled as a Graduate student, and may not reflect new departmental names following university-wide restructuring]) also used my template to solicit feedback in their own guest lecture activities.

Collectively, these experiences helped formalize my commitment to reflective teaching and ongoing pedagogical refinement.

5.0 Translating Hands-On Experience to the Classroom

Remote Sensing and GIS. Over the past six years, my primary professional focus has been on planetary science research that relies extensively on remote sensing and GIS methodologies. This work has required the management, organization, and analysis of large and complex datasets, often on the order of hundreds of gigabytes, drawn from a wide range of orbital and *in situ* sensors. Through this work, I have developed a deep, practical understanding of the strengths, limitations, and trade-offs inherent in different sensor systems and data products.

My research routinely involves the development of reproducible geospatial workflows and analytical pipelines using ArcGIS Pro, and custom statistical and machine learning approaches comparable to emerging GeoAI techniques[4]. These experiences have required not only technical proficiency, but also the ability to navigate real-world challenges related to data availability, acquisition constraints, preprocessing, uncertainty, and computational limitations; issues that students frequently encounter but are rarely emphasized in traditional ‘textbook’ examples.

Project Management. In addition to my technical expertise, I bring substantial project management experience to my teaching. I have led research projects to completion and work in collaboration with colleagues from Canada, the US, UK, and across Europe.

While recent years have been research-intensive, the extensive hands-on experience will directly inform my instructional approach. I understand the practical challenges faced by professionals working in remote sensing and GIS, and I will intentionally bring this perspective into the classroom. My teaching will emphasize connecting theoretical concepts to real-world applications, framing technical exercises around authentic problems, and helping students develop transferable skills in data management, workflow design, and analytical decision-making. By drawing on my applied research background, I aim to prepare students not only to succeed in coursework, but to confidently apply GIS and remote sensing techniques in research, industry, and professional practice.

6.0 Reflective Summary

I began my journey into the scholarship of teaching and learning at the University of Calgary, where, under the guidance of experienced facilitators and faculty, my teaching philosophy evolved into a deliberate, evidence-informed pedagogical practice. Across classroom, laboratory, field, and non-academic teaching contexts, I have come to understand teaching as a reciprocal and relational process, one that is strengthened through dialogue, reflection, and responsiveness to learners’ diverse experiences and needs. **Throughout this dossier, a consistent set of commitments emerges**: fostering active and applied learning environments; emphasizing inquiry-led analysis and transferable skills over rote mastery; intentionally reducing structural, cultural, and socioeconomic barriers to education; and practicing transparent, purposeful pedagogy grounded in reflection and

[4] For example: Barrett, A. M., Balme, M. R., **Favaro, E. A.**, Woods, M. J., Malinowski, M., Elsarky, L., … & Patel, M. R. (2026). Detecting and measuring Transverse Aeolian Ridges on Mars using deep learning. *Icarus*, 116947.

feedback. My experiences as a researcher, mentor, and scientific leader have reinforced the importance of connecting theory to real-world problems and equipping students with transferable analytical and professional skills. As I continue to develop as an educator, **I remain committed to ongoing pedagogical growth, inclusive and reflective practice, and creating learning environments** that empower students not only to succeed academically, but to engage critically, confidently, and responsibly with the complex challenges they will encounter beyond their university classrooms.

Appendix A: Guest Lecturer Feedback Form

Term: XXXXX
Course: XXXX

Guest Lecturer Evaluation

[First Name, Last Name]

This comment sheet is useful in providing feedback to the Guest Lecturer, **[First Name, Last Name]**. Your constructive comments will be taken into consideration for future planning of lecture material. *<u>You are not required to submit an evaluation. This is done on a volunteer basis, for the sole purpose of improving Graduate Student Teaching. There will be no repercussions for declining to provide feedback.</u>* Thank you for taking the time to complete this evaluation to the best of your knowledge.

Please select the lectures you attended which were taught by the Guest Lecturer

☐ Lecture Title and Course Date

What is your Major?

☐ Geography

☐ Environmental Science

☐ Geology

Other: ________________________

This course, [COURSE IDENTIFIER] , is a requirement of my degree

☐ Yes ☐ No

To date, I have attended:

100% of Classes | 99-90% | 89-80% | 79-70% | 69-60% | 59-50% | <50%

In this class, I expect to receive a grade of:

100% | 99-90% | 89-80% | 79-70% | 69-60% | 59-50% | <50%

SA - strongly agree; **A -** agree; **D** - disagree; **SD** - strongly disagree

Guest Lecturer - GL

Presentation of Material	SA	A	D	SD
The GL's presentation(s) was relevant to the concepts taught in [COURSE CODE]				
The concepts presented were communicated clearly				
The use of materials (PowerPoint) was helpful in following along with the GL				
When appropriate, the GL encourages student's participation				
The GL provided time during the lecture or at the end of lecture for questions				
The use of summaries throughout or at the end of lecture were useful				
Delivery of Presentation				
Relates current course content to what has been presented before/will be presented				
Makes course content relevant with references to "real world" applications				
Explicitly states relationships among various topics and facts/theory				
Explains difficult concepts or problems in more than one way				
Presents new content in more than one way (graphs, diagrams, text, examples)				
Presents up-to-date developments in the field				
Delivers content that is engaging				
Lecturer Feedback				
The GL was prepared for class				
The GL displays a clear understanding of course concepts				
The GL effectively communicated the subject matter of the lecture				
The GL spoke clearly and was easy to understand				
I am receptive to the teaching style of the GL				
Additional Comments:				

Appendix B: Guest Lecture Feedback Comments

Fall 2018, Geography 437, 11 Respondents

- "All the guest lectures for this course have been great, and this one is no exception. The lecture was clearly very well thought out and I appreciate you taking the time to prevent this our class + all the thought you put into it!"
- "Really cool presentation! Yay! Planetary Mapping! So cool!"
- "Great enthusiasm, really good job keeping the class engaged with discussion & questions, interesting hands-on activity to end class"

Presentation of Material	SA	A	D	SD
The GL's presentation(s) was relevant to the concepts taught in GEOG 407	**8**	**3**	**0**	**0**
The concepts presented were communicated clearly	**11**	**0**	**0**	**0**
The use of materials (PowerPoint) was helpful in following along with the GL	**10**	**1**	**0**	**0**
When appropriate, the GL encourages student's participation	**3**	**7**	**0**	**0**
The GL provided time during the lecture or at the end of lecture for questions	**10**	**1**	**0**	**0**
The use of summaries throughout or at the end of lecture were useful	**7**	**3**	**0**	**0**
Delivery of Presentation				
Relates current course content to what has been presented before/will be presented	**6**	**5**	**0**	**0**
Makes course content relevant with references to "real world" applications	**9**	**2**	**0**	**0**
Explicitly states relationships among various topics and facts/theory	**6**	**5**	**0**	**0**
Explains difficult concepts or problems in more than one way	**3**	**8**	**0**	**0**
Presents new content in more than one way (graphs, diagrams, text, examples)	**8**	**3**	**0**	**0**
Presents up-to-date developments in the field	**10**	**1**	**0**	**0**
Delivers content that is engaging	**8**	**3**	**0**	**0**
Lecturer Feedback				
The GL was prepared for class	**11**	**0**	**0**	**0**
The GL displays a clear understanding of course concepts	**11**	**0**	**0**	**0**
The GL effectively communicated the subject matter of the lecture	**10**	**1**	**0**	**0**
The GL spoke clearly and was easy to understand	**10**	**1**	**0**	**0**
I am receptive to the teaching style of the GL	**11**	**0**	**0**	**0**

SA - strongly agree; **A -** agree; **D** - disagree; **SD** - strongly disagree; **Guest Lecturer** - GL

Winter 2017, Geography 407, 4 Respondents

- "You can tell she's really enthusiastic about the material and is really knowledgeable about the content + can answer any question posed at her. I love her summary slides too, if you miss something during the lecture or not quite get something, the summaries really help with that. I think it can be beneficial to engage the audience even more, sometimes its hard to pay attention to everything for an hour. I liked the guessing which planet slides for the solar system lecture! Overall, I really like her lecture style and feel like I learn a lot of cool, interesting new material in her lectures!"
- "Excellent presentations, clearly discussed concepts and shared interesting insight into topics"
- "Engaging, friendly, enthusiastic – makes the subject interesting and is very knowledgeable"

Presentation of Material	SA	A	D	SD
The GL's presentation(s) was relevant to the concepts taught in GEOG 437	**2**	**2**	**0**	**0**
The concepts presented were communicated clearly	**4**	**0**	**0**	**0**
The use of materials (PowerPoint, maps, videos, interactive content) was helpful in following along with the GL	**3**	**1**	**0**	**0**
When appropriate, the GL encourages student's participation	**4**	**0**	**0**	**0**
The GL provided time during the lecture or at the end of lecture for questions	**4**	**0**	**0**	**0**
Delivery of Presentation				
Relates current course content to what has been presented before/will be presented	**1**	**2**	**1**	**0**
Makes course content relevant with references to "real world" applications	**3**	**1**	**0**	**0**
Explicitly states relationships among various topics and facts/theory	**3**	**1**	**0**	**0**
Explains difficult concepts or problems in more than one way	**1**	**3**	**0**	**0**
Presents new content in more than one way (graphs, diagrams, text, examples)	**3**	**1**	**0**	**0**
Presents up-to-date developments in the field	**4**	**0**	**0**	**0**
Delivers content that is engaging	**4**	**0**	**0**	**0**
Lecturer Feedback				
The GL was prepared for class	**4**	**0**	**0**	**0**
The GL displays a clear understanding of course concepts	**4**	**0**	**0**	**0**
The GL effectively communicated the subject matter of the lecture	**3**	**1**	**0**	**0**
The GL spoke clearly and was easy to understand	**4**	**0**	**0**	**0**
I am receptive to the teaching style of the GL	**4**	**0**	**0**	**0**

SA - strongly agree; **A -** agree; **D** - disagree; **SD** - strongly disagree; **Guest Lecturer** - GL

Appendix C: Additional Formal Feedback

UNIVERSITY OF CALGARY

Graduate Assistant Performance Review

Graduate Assistant Name: Favaro, Elena UCID: 10154384

[redacted] Department : Geography

Category of Appointment: Full GAT - Geog 403 Semester Period of Appointment: Winter 2018

Date of Performance Review: May 5, 2018

Purpose:
As stated by the Collective Agreement, written performance feedback is encouraged to ensure Graduate Assistants are developing successful professional skills and have guided feedback to improve their overall performance.

Rating Scale Defined:

E	Exceptional
S	Satisfactory
U	Unsatisfactory
N	Unknown / Not Applicable

General Evaluation:

Evaluation Areas	Rating
1. GA effectively completes assignments according to the Assignment of Assistantship Duties form and /or project requirements as scheduled.	E
2. GA is reliable and prompt, consistently demonstrating accountability for scheduled office/laboratory/lecture time.	E
3. GA is proficient and professional in oral and written communications and communicates effectively to foster and promote academic achievement.	E
4. GA demonstrates technical and functional knowledge.	E
5. GA interacts with students and supervisor effectively, contributing to an inclusive and welcoming learning environment.	E
6. GA develops trust and demonstrates ethical behavior, including following all regulations regarding confidential information.	E
7. GA appropriately handles issues and conflicts by analyzing problems effectively.	E

Evaluators Comments:
(Please comment on specific aspects of the assistant's general performance by highlighting strengths and areas of development.)

Elena did an excellent job this semester. She worked independently, and delivered all of the lab assignments without any problems. Assignments were marked in a reasonable amount of time, and I basically didn't hear anything from the students. Typically that's a very good sign, because if things are not going well in lab the students are usually pretty quick to bring their issues to me. Elena made my job a lot easier this semester, and my only regret is that I probably won't get another chance to work with her.

Graduate Assistant's Comments:
(Please comment on specific aspects of the supervisor's assignments and guidance.)

Working with Brent this semester has been a wonderful end to my Teaching Assistant career. I am so grateful for his guidance and support.

Brent allowed me the freedom to deliver labs in a manner I thought most effective for our students. I was also consulted and brought in on the re-imaging of current labs, for which I am grateful. Brent is always looking for ways to improve the student experience, and this year that included additional presentations from students and a "flipped classroom" exercise, where students were given the opportunity to receive anonymous feedback on their final research paper from others in the class.

When I did require clarification or direction, Brent was quick to make time to meet with me. We collaborated extremely well, and I'm disappointed I won't have another opportunity to work with him in this capacity again.

The signature of the Graduate Assistant and Instructor of Record/Researcher indicates that they have had the opportunity to review and discuss the Assistant's performance; it does not imply agreement.

Graduate Assistant: ______________ ______________ ______________

Print Name | Signature | Date

Instructor of Record/Researcher: ______________ ______________ ______________

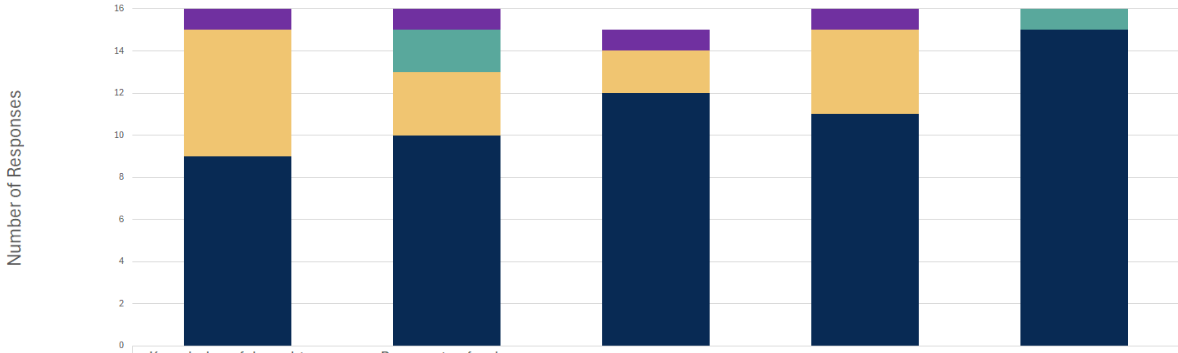
Print Name | Signature | Date

Cc: Student Copy
Department File

Winter 2015, Geography 403, 16 Total Respondents

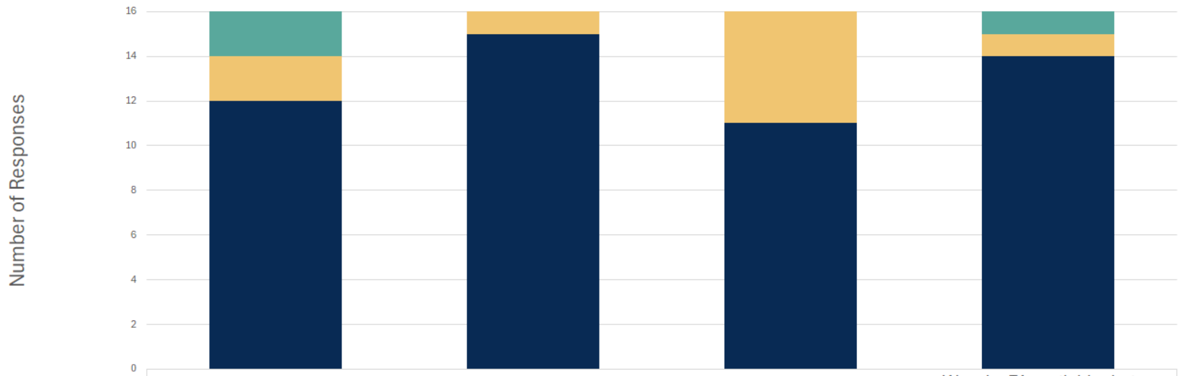
Summary of feedback

Please rate your Teaching Assistant's ability to teach the labratory compoent of this course



	Knowledge of the subject area	Preparation for the laboratory exercise	Teaching skills	Clarity of communication	Enthusiasm
Unsatisfactory	0	0	0	0	0
Below Average	1	1	1	1	0
Average	0	2	0	0	1
Above Average	6	3	2	4	0
Very Good	9	10	12	11	15

Please comment on the student interaction skills shown by your Teaching Assistant



	Fair evaluation	Approachability	Assignments returned within a reasonable time	Was the TA available during SCHEDULED labs and office hours. Note that Tas are not required to be available outside these times
Unsatisfactory	0	0	0	0
Below Average	0	0	0	0
Average	2	0	0	1
Above Average	2	1	5	1
Very Good	12	15	11	14

Summary of Additional Comments

“Friendly and approachable, great with excel software and database files”

“Coolest TA I’ve ever had!”

“Super approachable and funny”

“She was great and often went beyond the scope of her TA expectations and promoted skills and resources that would help us beyond this class”

“Elena is a very approachable and fair in her marking/handling of students (not too easy on them. I expect an A.”

Elena approaches this position with uncommon enthusiasm. She has been very available for additional help and provides constructive feedback. One of the best T.A.s I’ve had!”

“Fantastic TA. Best I’ve had at U of C. She should get an award for sure! Super helpful and approachable.”

“Great job, very good TA”

Very good 1 on 1, lots of help with excel.”

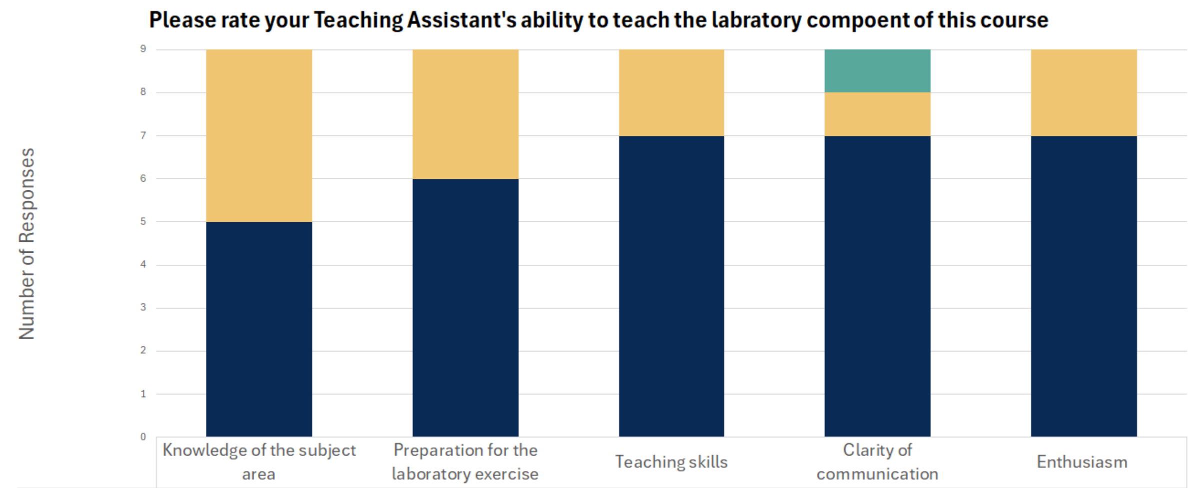
“Kick ass”

“Was difficult in marking; however she went out of her way to help students and offered a lot of out of class time/lab time assistance”.

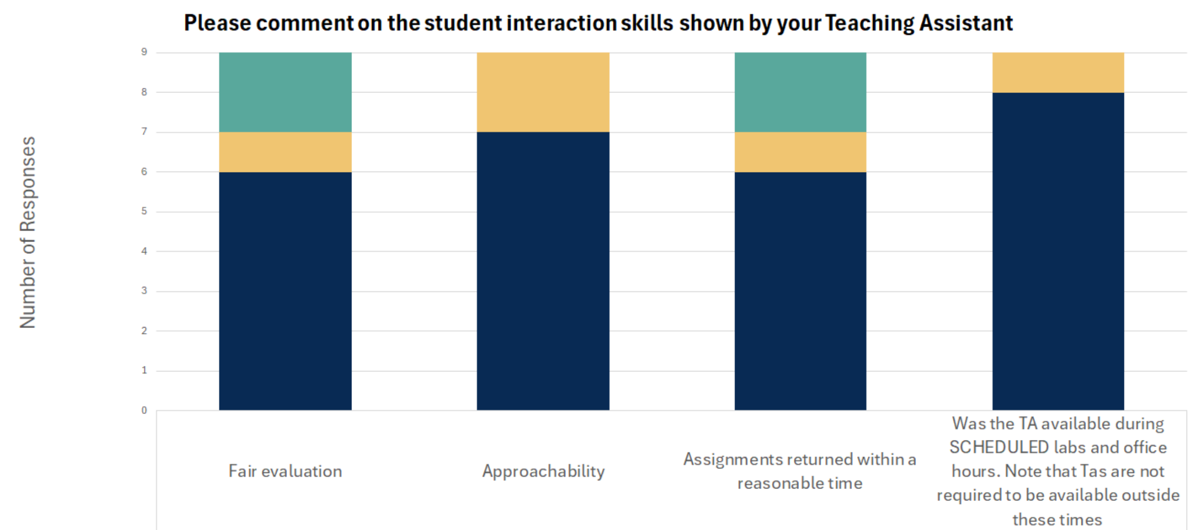
“Elena is a great T.A. Taugh[t] material with enthusiasm. Good knowledge of material and was always prepared for lab sessions”.

Winter 2016, Geography 403, 9 Total Respondents

Summary of feedback



	Knowledge of the subject area	Preparation for the laboratory exercise	Teaching skills	Clarity of communication	Enthusiasm
Unsatisfactory	0	0	0	0	0
Below Average	0	0	0	0	0
Average	0	0	0	1	0
Above Average	4	3	2	1	2
Very Good	5	6	7	7	7



	Fair evaluation	Approachability	Assignments returned within a reasonable time	Was the TA available during SCHEDULED labs and office hours. Note that Tas are not required to be available outside these times
Unsatisfactory	0	0	0	0
Below Average	0	0	0	0
Average	2	0	2	0
Above Average	1	2	1	1
Very Good	6	7	6	8

Summary of Additional Comments

“Without a doubt, one of the very best T.A.s in the department. Would highly recommend her for any T.A. awards. Will try to take courses with her again”.

“Labs should be 100% error free, no spelling errors”

“Easily one of the best TA’s I’ve had in the past 5 years!”

“Elena is a wonderful TA. She wants the best for her students and helps in ensuring everyone is doing their best work”.

“Some of the labs were a little too tough”.

“She’s the best TA I’ve ever had”.

Appendix D: Optional Assignment for Geography 403

This version of the lab was the first introduced in 2015

Geography 403: Physical Oceanography and Climate Variability

Optional Lab 5: Skeptical Science and Climate Change

Introduction

The Intergovernmental Panel on Climate Change (IPCC) has been saying for years that the climate is changing due to anthropogenic forcing. Let that sink in for just a moment. Since the start of the Industrial Revolution, temperatures have changed (and in many cases, risen) in unnatural ways. In 165 years, such dramatic changes to our weather and climate have taken place that many scientists have cautioned we're coming up to the point of no return. However, despite the vast body of evidence to support these claims, there are groups of people who deny a) the climate is changing at all and b) human are at least partially at fault for this change.

This type of skepticism is alive and well in science, and in fact plays an important role in the development of theories, the improvement of techniques, and the debunking of erroneous claims. For example, in the late 1990's, Andrew Wakefield (formally Dr. Andrew Wakefield, a British surgeon and medical researcher), published a paper linking the rise in childhood Autism with the early childhood administration of the MMR (Measles, Mumps, and Rubella) vaccine. His claims caught on with the public, who were eager for an answer into the alarmingly high rates of Autism. Jenny McCarthy, an American celebrity, public ally linked her son's Autism with his childhood vaccinations and called for a stop to the vaccination of children. What Ms. McCarthy and those who agreed with her did not realize is that Mr. Wakefield utterly falsified his data. This breach of ethics was so severe that following the retraction in the journal The Lancet, where the study was originally published, the UK Medical Register revoked Mr. Wakefield's medical license. Further investigation into Mr. Wakefield's work showed that he himself was working on a new MMR vaccine, one that he hoped would be a lucrative option for parents, doctors, and public alike, who were clamouring for a less "harmful" vaccine option.

This is just one example of how scientific knowledge was distorted by one false claim, and how these ideas still persist today, despite the overwhelming evidence against Mr. Wakefield's statements. This is what Climate scientists face.

Assignment

This assignment will (hopefully) help you to develop a more critical eye when consuming media information on Climate Change. Not every denier is a drummed-up politician, and not every expert has the ability to disseminate complex climate information to the public.

For this assignment, you are asked to research how the media has contributed to the skepticism around climate change. The articles/videos you read/watch must be from reputable news sources (CBC, NBC, Daily Newspapers, videos which are recorded as part of official record [i.e. CSPAN or CPAC]).

The write-up for this assignment must not exceed 4 pages (Optional title page and mandatory references not included, 1.5 spacing, 12pt font, "normal" margins [a margin setting in word]) and must contain the following aspects:

- You must choose at least 3 different news sources, with a balance between print and multimedia.
- *Briefly* synopsize the content of each source
- Critically examine the claims being made and critique them accordingly.
    - Some things to consider would be:
        - Is their claim based in fact?
        - Do they understand the terminology they are using?
        - Do they perhaps have ulterior motives for denying change?

This Assignment is to be handed in **No Later than 11:59MT on the 11th of April.** I will not accept late submissions sent to my email. If I do not receive an assignment by this time, I will assume you have opted not to do it.